



Sliced Optimal Transport

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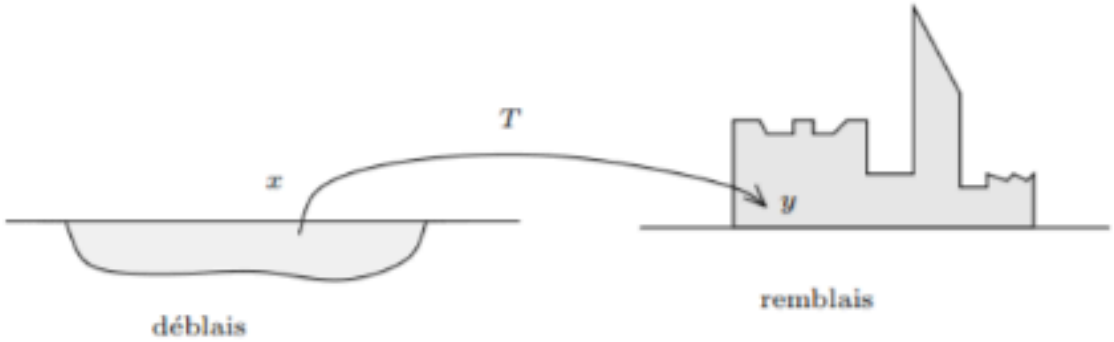
Optimal Transport for Machine Learning

October 29, 2025

Reminders on Optimal Transport

The First OT Problem (Monge, 1781)

How do you move sand into a pit at minimal cost?



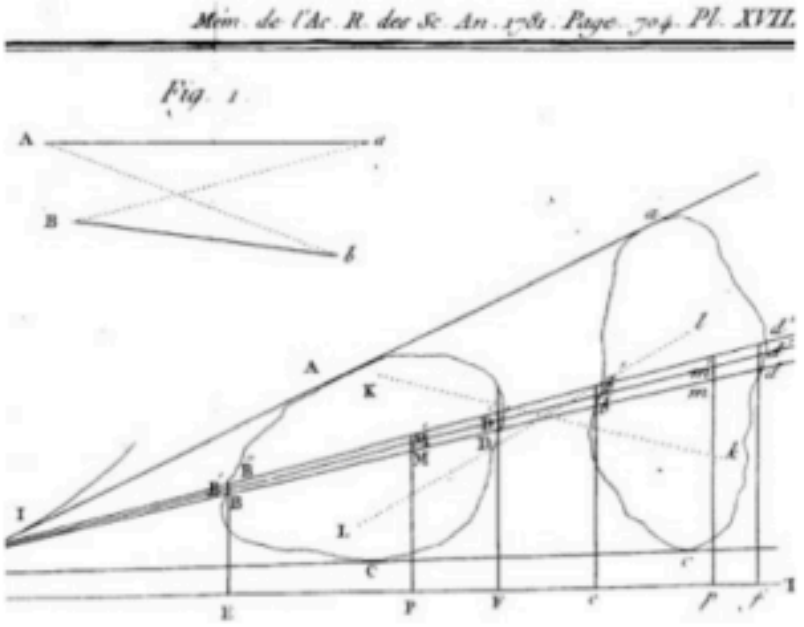


MÉMOIRE
SUR LA
THÉORIE DES DÉBLAIS
ET DES REMBLAIS.

Par M. MONGE.

Lorsqu'on doit transporter des terres d'un lieu dans un autre, on a coutume de donner le nom de *Déblai* au volume des terres que l'on doit transporter, & le nom de *Remblai* à l'espace qu'elles doivent occuper après le transport.

Le prix du transport d'une molécule étant, toutes choses d'ailleurs égales, proportionnel à son poids & à l'espace qu'on lui fait parcourir, & par conséquent le prix du transport total devant être proportionnel à la somme des produits des molécules multipliées chacune par l'espace parcouru, il s'ensuit que le déblai & le remblai étant donnés de figure & de position, il n'est pas indifférent que telle molécule du déblai soit transportée dans tel ou tel autre endroit du remblai, mais qu'il y a une certaine distribution à faire des molécules du premier dans le second, d'après laquelle la somme de ces produits sera la moindre possible, & le prix du transport total sera un minimum.



Mémoire sur la théorie des déblais et des remblais, Monge, 1781.

A Versatile Tool for Comparing Distributions



Assignment between point clouds

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν

A Versatile Tool for Comparing Distributions



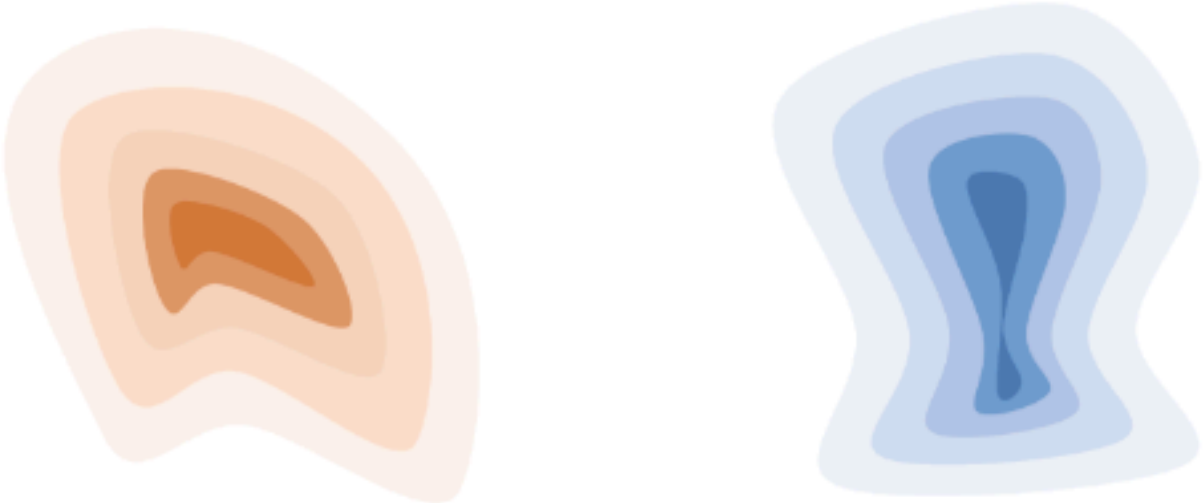
Transport between discrete distributions

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν

A Versatile Tool for Comparing Distributions



Transport between continuous distributions

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν

A Versatile Tool for Comparing Distributions



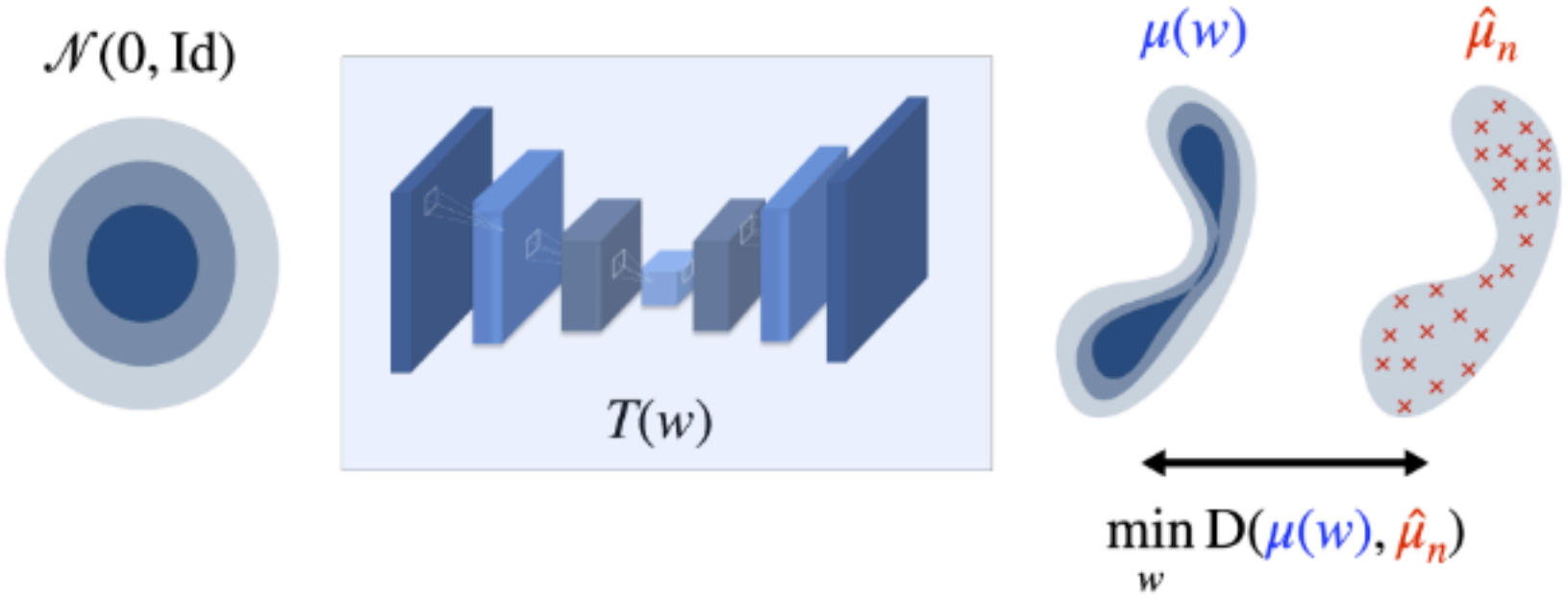
Transport between continuous/discrete distributions

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν

OT for Machine Learning



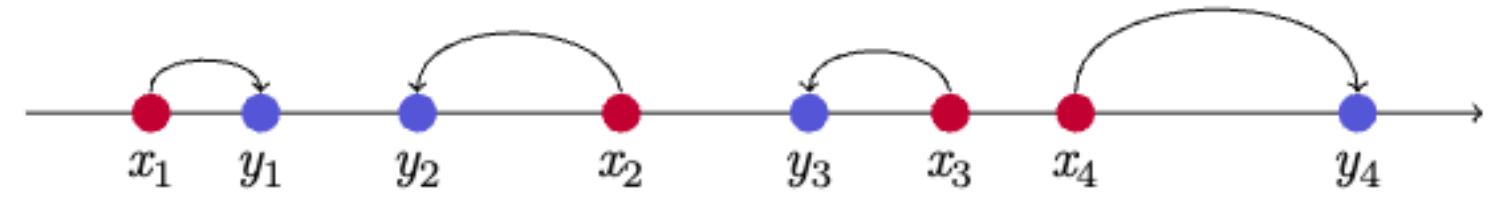
Generative models

Some OT-based deep generative models:

- Wasserstein GANs [Arjovsky et al., 2017]: GANs with objective function based on the dual of $\mathbf{W}_1(\mu(w), \hat{\mu}_n)$
- Wasserstein Auto-Encoders [Tolstikhin et al., 2018]: Auto-encoders trained by minimizing a regularized $\mathbf{W}_1(\mu(w), \hat{\mu}_n)$

OT in One Dimension: Sort and Match

Let $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\nu = \frac{1}{n}\sum_{i=1}^{n} y_i$ with $x_i, y_i \in \mathbb{R}$



source (μ) target (ν)

$$\mathbf{W}_2^2(\mu, \nu) = \frac{1}{n}\sum_{i=1}^{n} |x_{(i)} - y_{(i)}|^2$$

with $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$, and $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(n)}$

Fast to compute: Just sort $\rightarrow \mathcal{O}(n\log(n))$

OT in Multi-Dimensions

Now, μ, ν are distributions on $\mathbb{R}^d$

Discrete measures

Let $\mu_n = \sum_{i=1}^n a_i \delta_{x_i}$ and $\nu_n = \sum_{j=1}^n b_j \delta_{y_j}$

$$\mathbf{W}_2^2(\mu_n, \nu_n) = \min_{\pi \in \Pi(a,b)} \left\{ \langle C, \pi \rangle \triangleq \sum_{i=1}^n \sum_{j=1}^n C_{ij} \pi_{ij} \right\}$$

with $C \in \mathbb{R}^{n \times n}$ the cost matrix s.t. $C_{ij} = c(x_i, y_j)$, and

$$\Pi(a, b) \triangleq \{\pi \in \mathbb{R}_+^{n \times n} \text{ s.t. } \pi \mathbb{1}_n = a, \ \pi^T \mathbb{1}_n = b\}$$

OT is a linear program

- Linear program with n^2 variables and $2n$ constraints
- Classical solvers: Hungarian algorithm, interior-point method
- Computational complexity $\mathcal{O}(n^3)$

Standard OT: Too Costly in ML

Computational bottleneck: Computing $\mathbf{W}_2(\mu_n, \nu_n)$ costs $\mathcal{O}(n^3)$

$\Rightarrow$ Expensive for modern ML pipelines

Example: Training OT-based neural networks

- Mini-batch $n = 256 \Rightarrow$ 1 evaluation of $\mathbf{W}_2(\mu_n, \nu_n) \sim 10^7$ operations
- Needed at every training step, over thousands of iterations

Standard OT is expensive $\Rightarrow$ Approximate/regularized OT

- Entropy-regularized OT (Sinkhorn algorithm) [Cuturi, 2013]
- Sliced OT [Rabin et al., 2011]

The Curse of Dimensionality

In practice, we only observe finite samples $(x_i)_{i=1}^n \overset{i.i.d.}{\sim} \mu$, $(y_i)_{i=1}^n \overset{i.i.d.}{\sim} \nu$

$$\mathbf{W}_p(\mu_n, \nu_n) \quad \text{vs.} \quad \mathbf{W}_p(\mu, \nu)$$

How fast does $\mathbf{W}_p(\mu_n, \nu_n) \to \mathbf{W}_p(\mu, \nu)$ as $n \to +\infty$?

Sample complexity of $\mathbf{W}_p$ [Dudley, 1969]

If μ, ν are supported on a bounded domain of $\mathbb{R}^d$, $d > 2$,

$$\mathbb{E}\big[\,|\mathbf{W}_p(\mu_n, \nu_n) - \mathbf{W}_p(\mu, \nu)|\,\big] = \mathcal{O}(n^{-\frac{1}{d}})$$

with $\mathbb{E}$ taken over $(x_i, y_i)_{i=1}^n$.

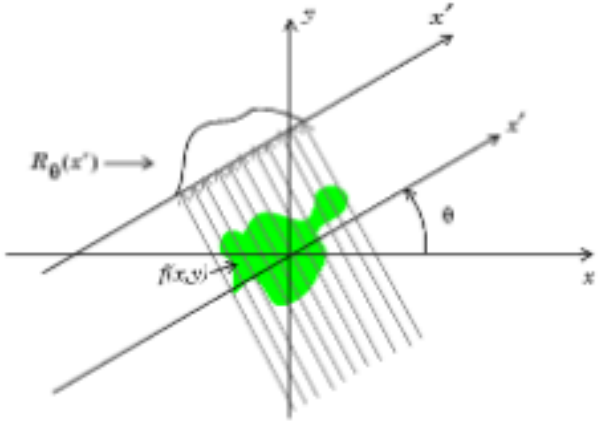
Consequences:

- Higher dimension $d \Rightarrow$ slower convergence
- Accurate estimates may require prohibitively large n

Sliced Optimal Transport

Connection to the Radon Transform

$\mathbf{SW}_p$ can be expressed with the Radon transform of measures

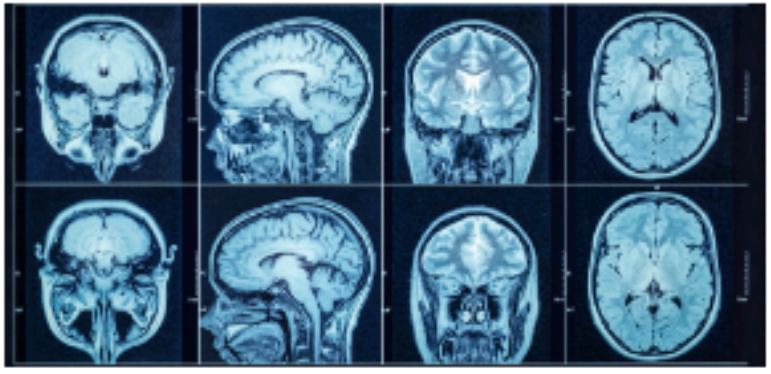


Radon transform (source)

Widely used for image reconstruction, *e.g.*, in tomography



CT scanner (source)

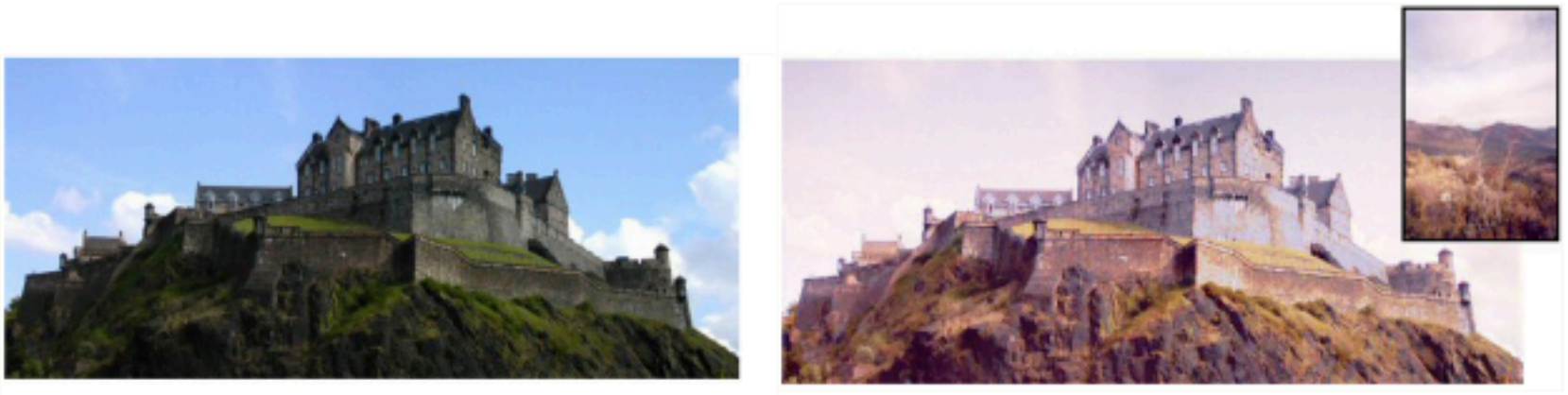


CT scans (source)

From Color Transfer...

Origins: Image processing

2006: Iterative Distribution Transfer (IDT): algorithm for color transfer, by iteratively matching 1D projections



Color transfer done with IDT [Pitié et al., 2007]

2012: Introduction of Sliced-Wasserstein distance (Bernot et al.)

2013: Theoretical analysis and link to gradient flows (Bonnotte's PhD thesis)

... To Generative Modeling



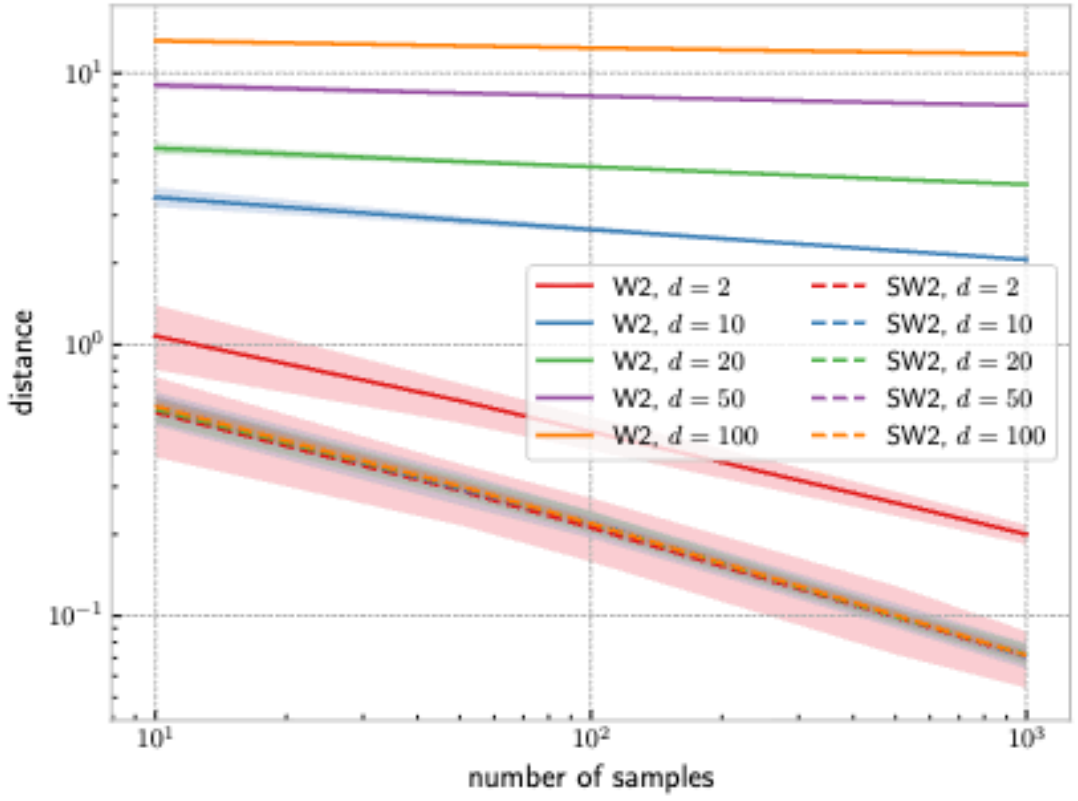
Images generated by (Sliced-)Wass. GANs with FID scores [Wu et al., 2019]

2017: Wasserstein GANs / auto-encoders

2018: Sliced-Wasserstein GANs / auto-encoders [Deshpande et al., 2018; Wu et al., 2019; Kolouri et al., 2019b]

2018 – present: Growing theory and applications of SW in ML

Slicing Beats the Curse of Dimensionality?



$\mathbf{SW}_2$ vs. $\mathbf{W}_2$ between two sets of n i.i.d. samples from $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$

Sample complexity of $\mathbf{SW}_p$

For μ, ν supported on a bounded domain of $\mathbb{R}^d$,

$$\mathbb{E}\big[\,|\mathbf{SW}_p(\mu_n, \nu_n) - \mathbf{SW}_p(\mu, \nu)|\,\big] = \mathcal{O}(n^{-\frac{1}{2p}})$$

$\Rightarrow$ Dimension-free rate, unlike $\mathbf{W}_p$ $(\mathcal{O}(n^{-\frac{1}{d}}))$

Still Some Practical Challenges!

The previous rate applies to $\mathbf{SW}_p(\mu,\nu) = \left[\mathbb{E}_{\theta\sim\mathcal{U}(\mathbb{S}^{d-1})}\mathbf{W}_p^p(\mu^\theta,\nu^\theta)\right]^{1/p}$, which is intractable in practice

Practical approximation uses k random projections:

$$\widehat{\mathbf{SW}}_p^p(\mu,\nu) = \frac{1}{k}\sum_{j=1}^{k}\mathbf{W}_p^p(\mu^{\theta_j},\nu^{\theta_j})$$

Monte Carlo approximation error

$$\mathbb{E}\left[\,\left|\widehat{\mathbf{SW}}_p^p(\mu,\nu) - \mathbf{SW}_p^p(\mu,\nu)\right|\,\right] \leq \frac{1}{k}\,\mathrm{Var}_\theta\left[\mathbf{W}_p^p(\mu^\theta,\nu^\theta)\right]$$

Trade-off: more projections $\Rightarrow$ better accuracy, but higher computational cost

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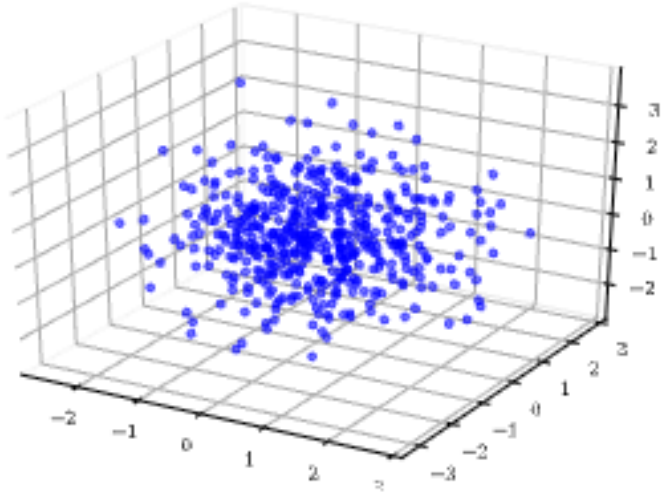
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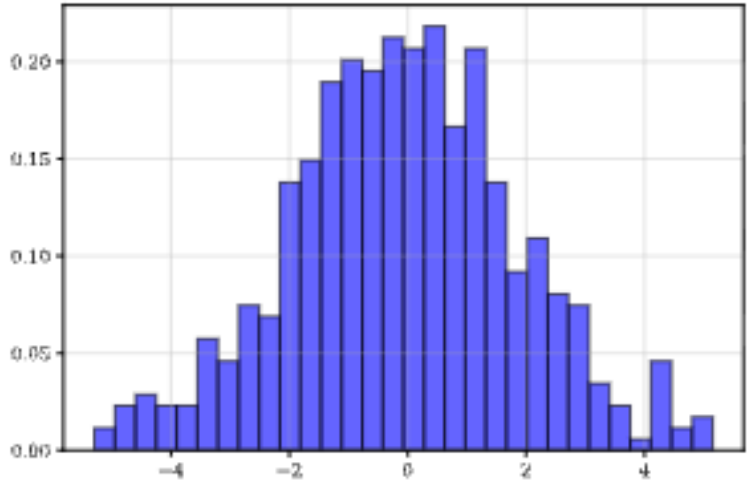
Slicing: From High to One Dimension

For $\theta = [\theta^{(1)}\, \theta^{(2)}\, \dots\, \theta^{(d)}]^\top \in \mathbb{R}^d$ and $x = [x^{(1)}\, x^{(2)}\, \dots\, x^{(d)}]^\top \in \mathbb{R}^d$, let

$$\langle \theta, x \rangle = \sum_{j=1}^{d} \theta^{(j)} x^{(j)} \in \mathbb{R}$$

project along $\theta \in \mathbb{R}^d$





$\mu = \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}$

$\mu^\theta = \frac{1}{n} \sum_{i=1}^{n} \delta_{\langle \theta, x_i \rangle}$